Mapping Elevation with Ordinary Kriging

Prediction and uncertainty surfaces for Michigan's Upper Peninsula

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Overview

Kriging uses the distance and direction between sample points to model spatial correlation and predict values at unsampled locations, along with the uncertainty of each prediction. This project used ordinary kriging in the ArcGIS Pro Geostatistical Wizard to build an elevation surface for Michigan's Upper Peninsula from elevation sample points, and paired it with a standard error surface showing where the predictions can be trusted.

Exploratory Analysis

Plotting elevation against X and Y coordinates showed only a very weak relationship ($R^2 = 0.14$). Once the axis minimum was set to zero, the slope became even less noticeable and the points remained widely dispersed, confirming that there was no strong overall trend to remove. The histogram and normal QQ plot indicated that a natural log transformation would help stabilize variance, so the data were log-transformed before modeling.

Model Setup

Spherical and circular semivariogram models were explored, and the model was optimized using cross-validation statistics. Anisotropy was enabled, because even small directional differences in spatial correlation affect both prediction accuracy and estimation variance. A lag size of 21,000 m was used, capturing interactions within a realistic distance while including enough neighboring points for a stable model.

Choosing the Search Neighborhood

Two search neighborhoods were compared, both with a maximum of 5 and a minimum of 2 neighbors: one sector and four sectors.

Cross-validation statistic	1 sector	4 sectors
Root mean square error	0.093	0.088
Mean standardized error	0.018	0.002
Root mean square standardized error	1.125	1.100
Average standard error	0.083	0.081

The four-sector model performed better on every statistic: lower errors, a mean standardized error closer to zero, and a root mean square standardized error closer to one. It better matches the semivariogram structure, reduces kriging variance and improves interpolation accuracy, so it was used for the final surface.

Results

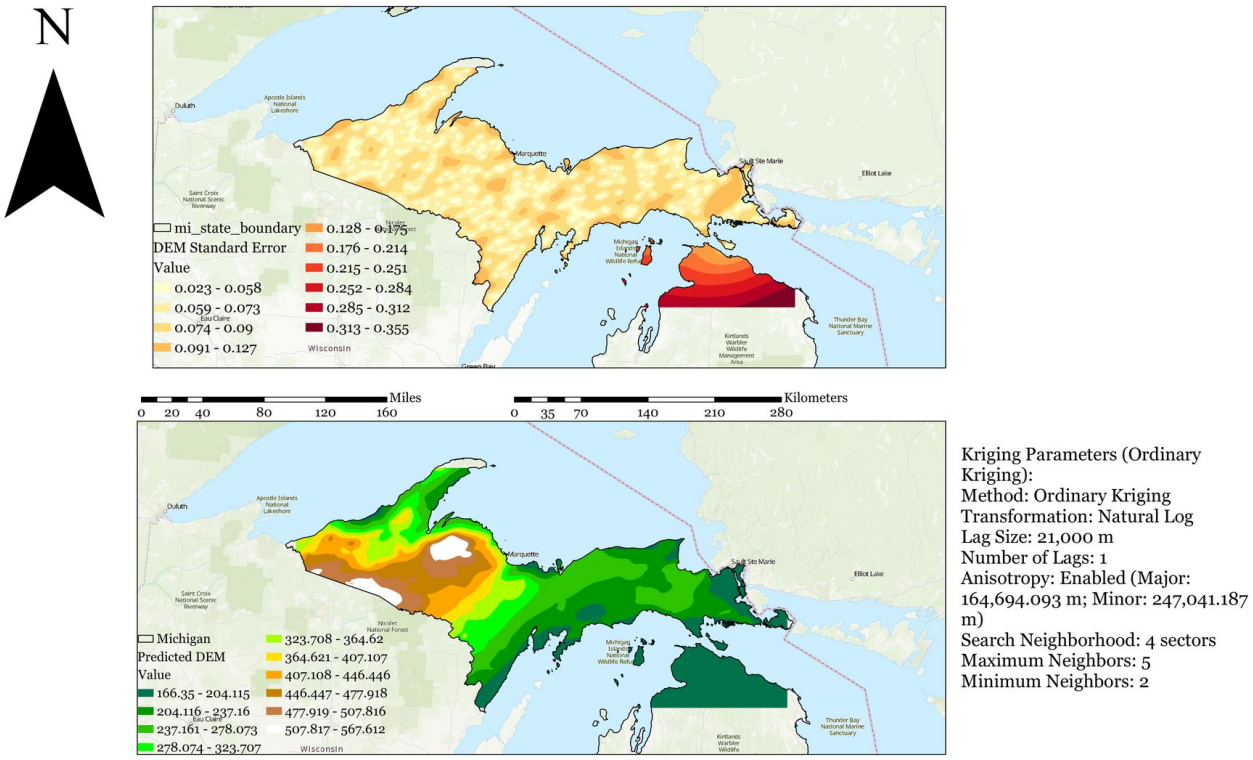


Figure 1. Kriging standard error (top) and predicted elevation (bottom) for the Upper Peninsula, with the model parameters used.

The prediction was exported as a 100 m raster, back-transformed from the log scale with the raster calculator, and clipped to the outline of the Upper Peninsula. The standard error surface shows the spatial pattern of uncertainty rather than elevation. Standard error is lowest where samples are dense, such as the central and western Upper Peninsula, because strong spatial autocorrelation and enough nearby points make predictions more reliable. It is highest near the boundaries and in sparsely sampled areas, particularly the southeast, where fewer neighbors and edge effects increase uncertainty.